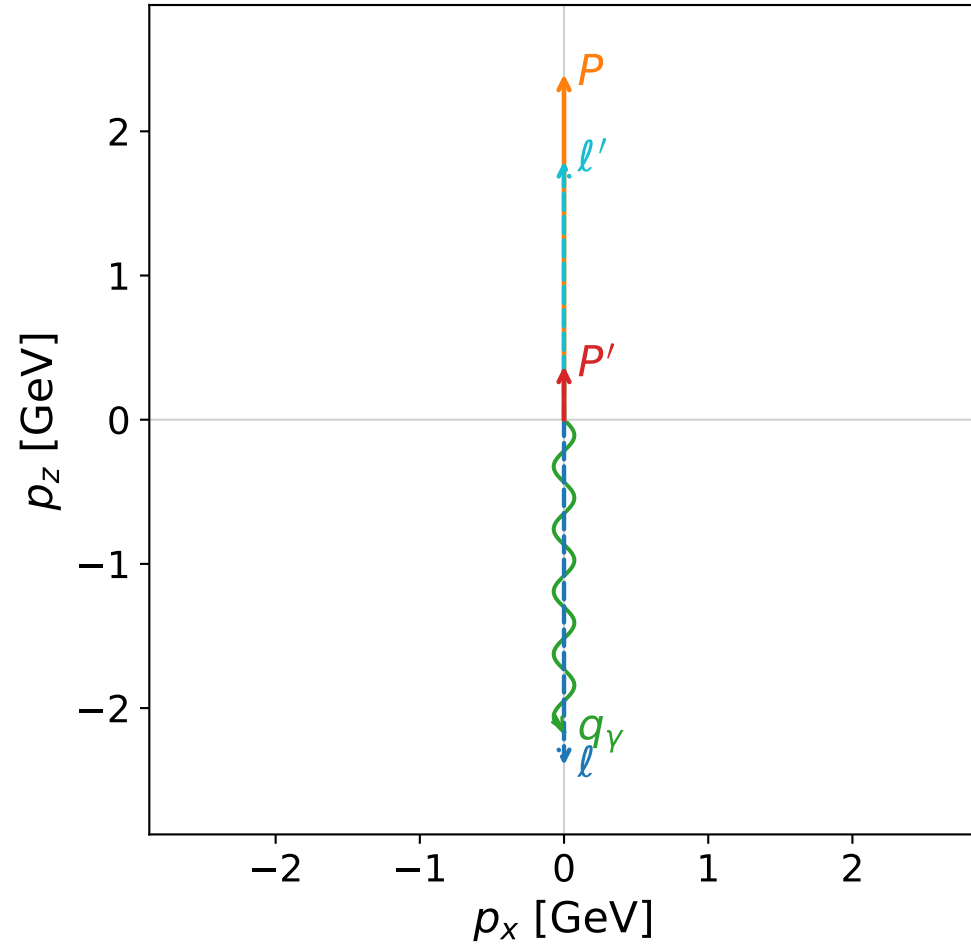


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

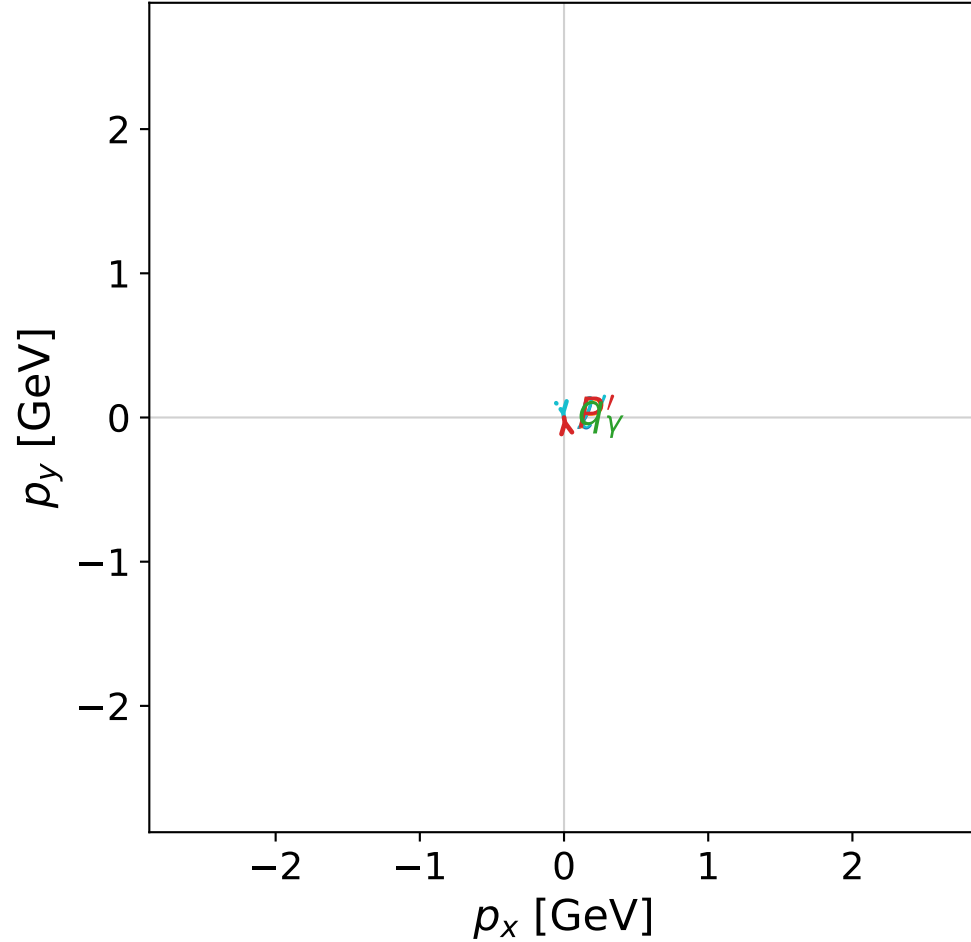


dghz_mixing_angles_polarization_02_local_minimum_0
 $d_{\text{GHZ}}=2.22835\text{e-}08$
 region: polarization_cluster_02_local_minimum_0
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.7579928$ rad
 incoming lepton: $0.726217|+\rangle + 0.687465|-\rangle$
 proton mixing angle: $\alpha_p=2.3835959$ rad
 incoming proton: $-0.726215|+\rangle + 0.687468|-\rangle$

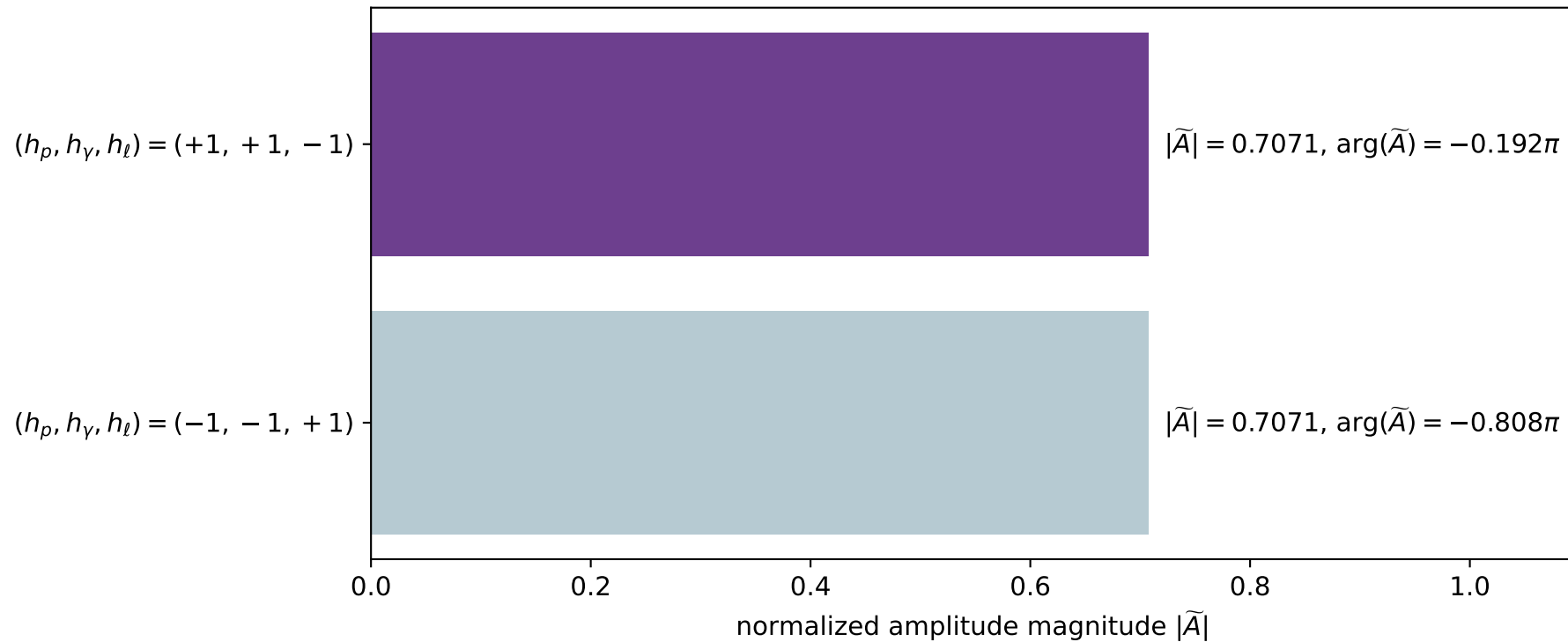
$s=24.7062$, $\sqrt{s}=4.97053$, $|\vec{P}|=2.39676$, $|\vec{P}'|=0.371823$
 $\theta_{p'}=1.30111\text{e-}07$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.74218$, $\phi_\gamma=3.74627$
 $E_\gamma=2.16667$, $\theta_{l'}=2.58096\text{e-}08$, $\phi_{l'}=4.88378$
 $Q^2=17.2073$, $x_B=0.741984$, $t=-1.65188$, $W^2=6.86346$, $y=0.973331$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3968$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3968$
 P $E= 2.5738$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3968$
 l' $E= 1.7948$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.7948$
 P' $E= 1.0090$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.3718$
 q_γ $E= 2.1667$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1667$

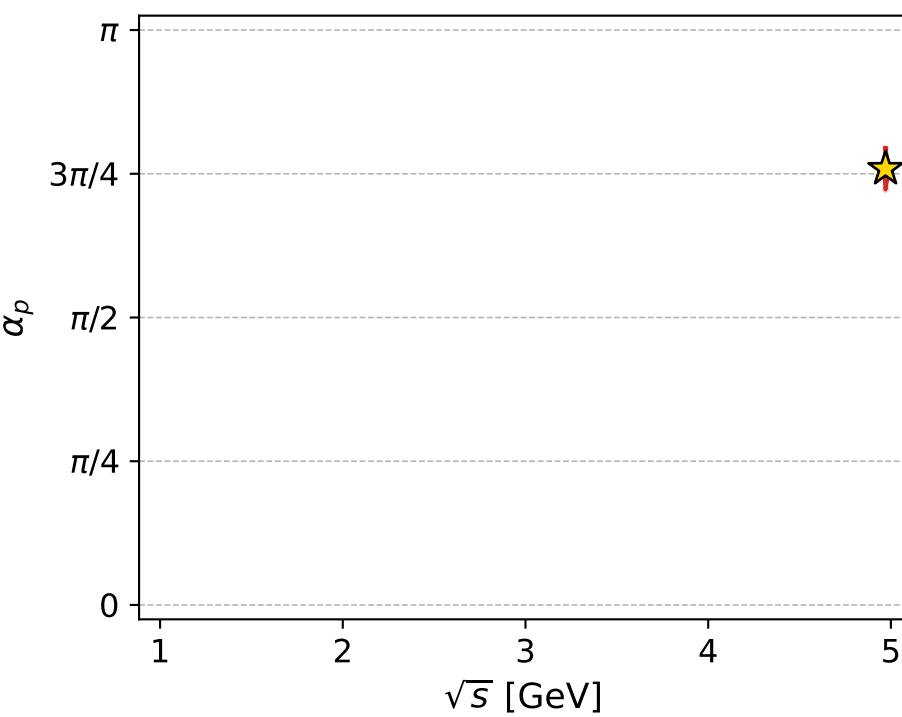
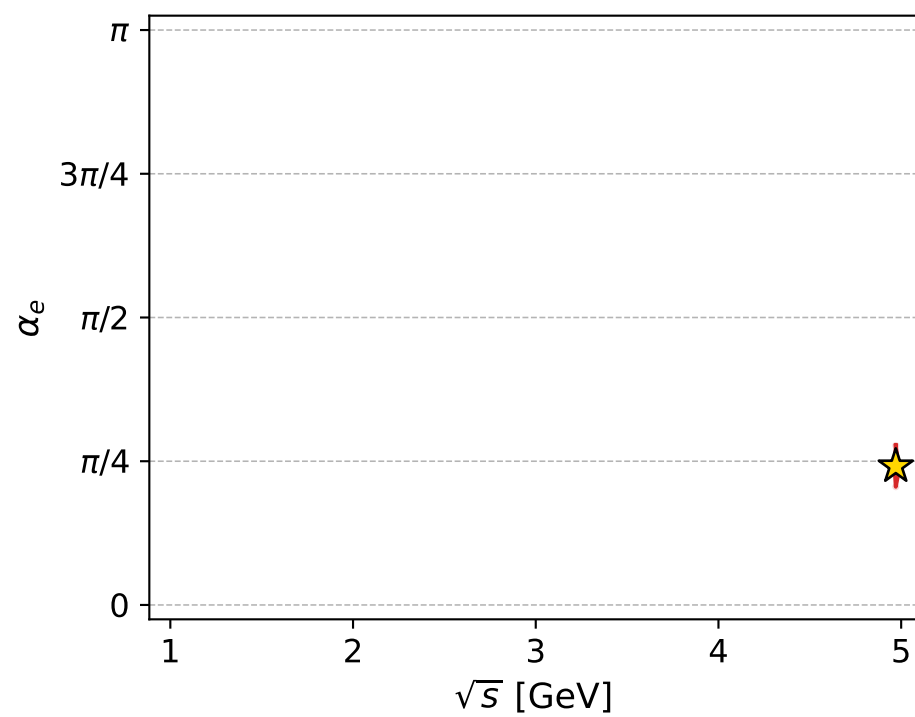
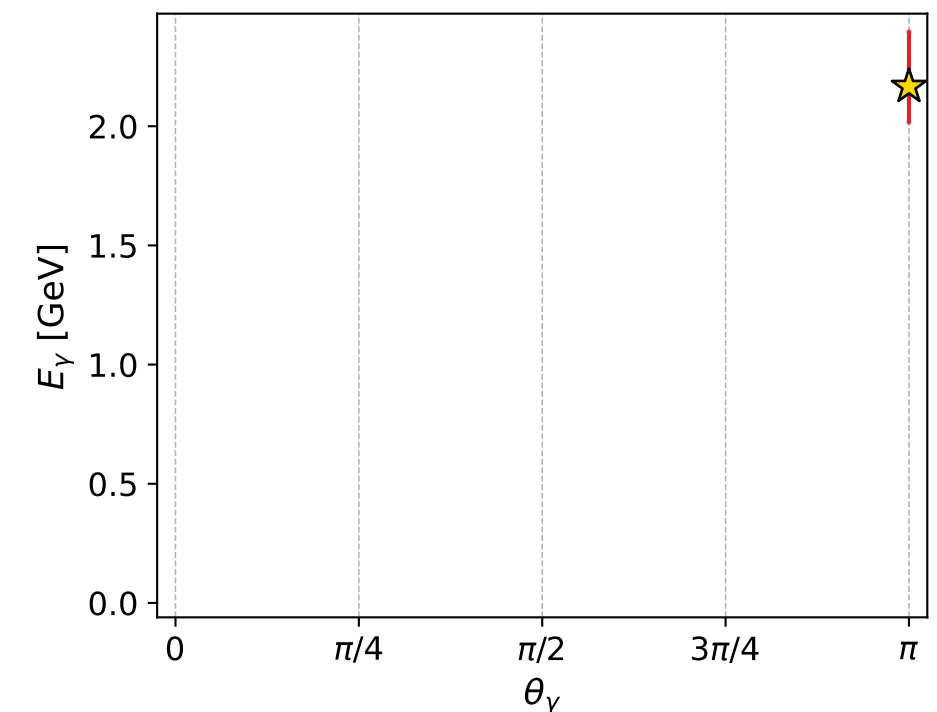
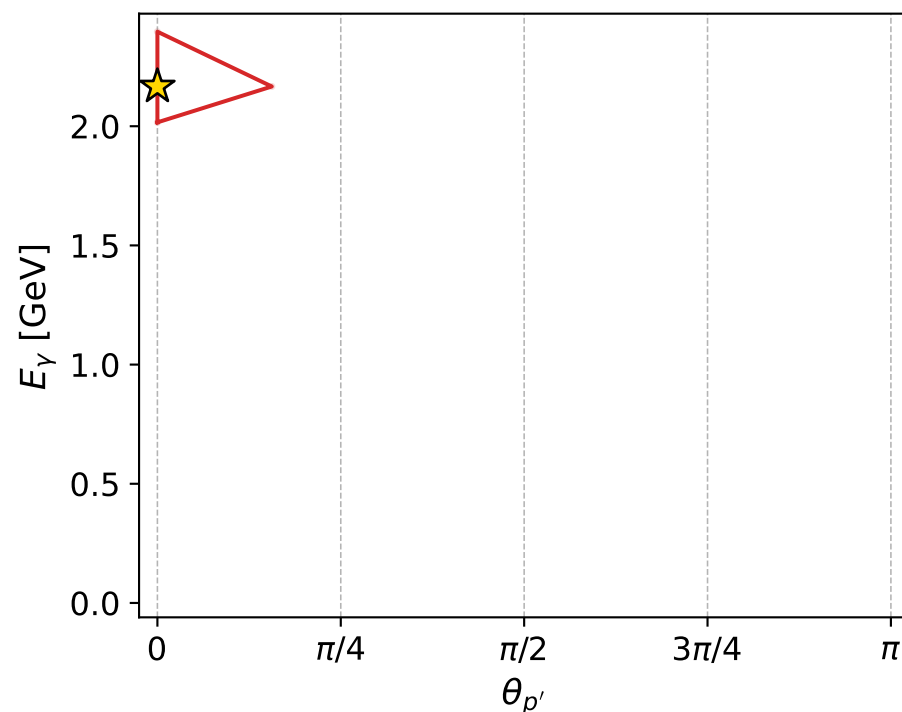
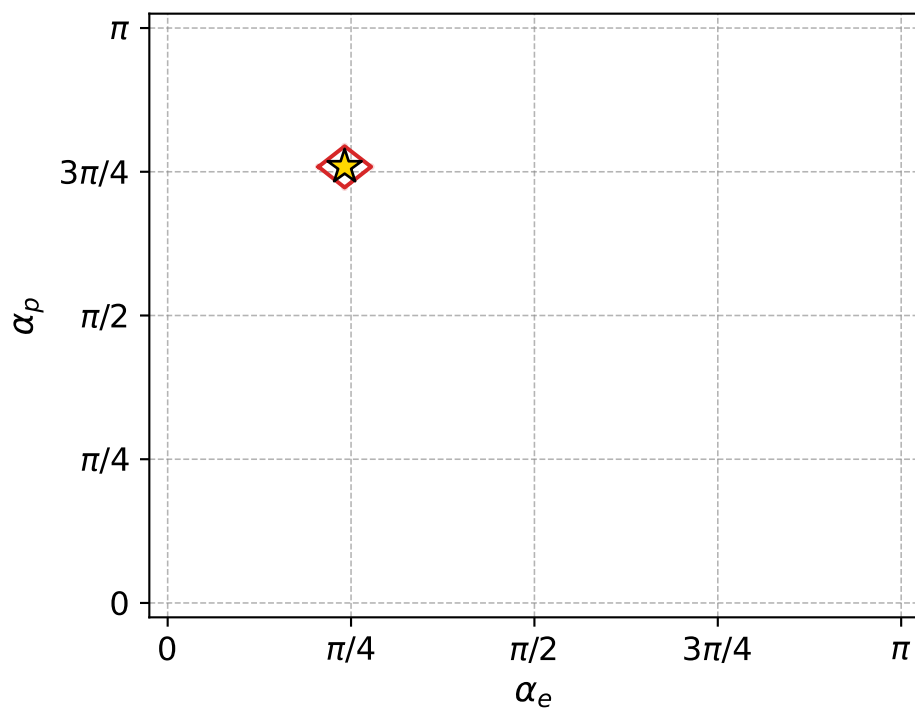
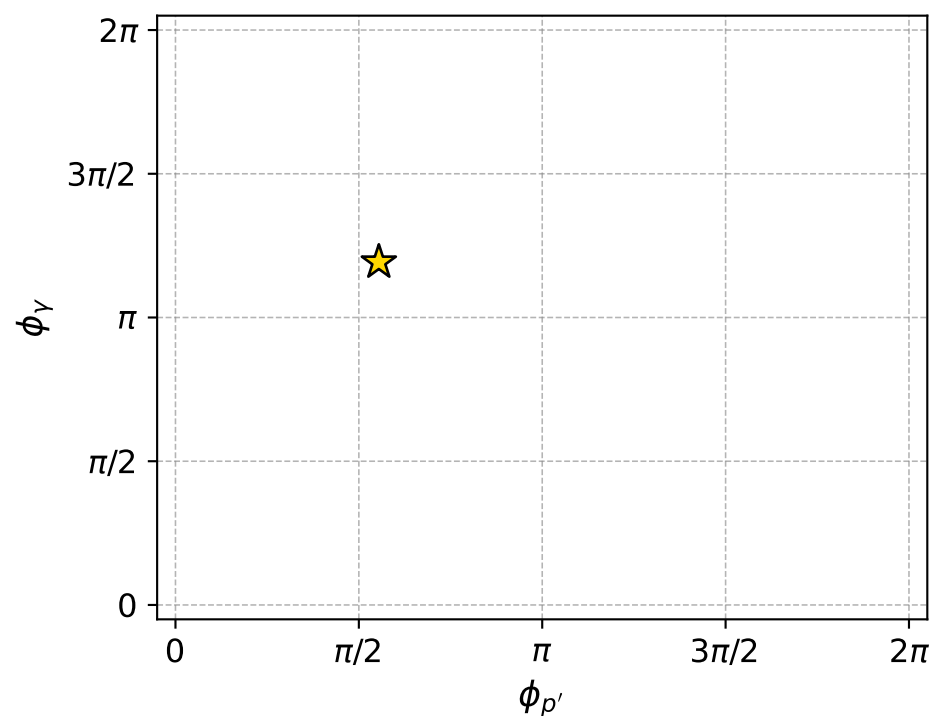
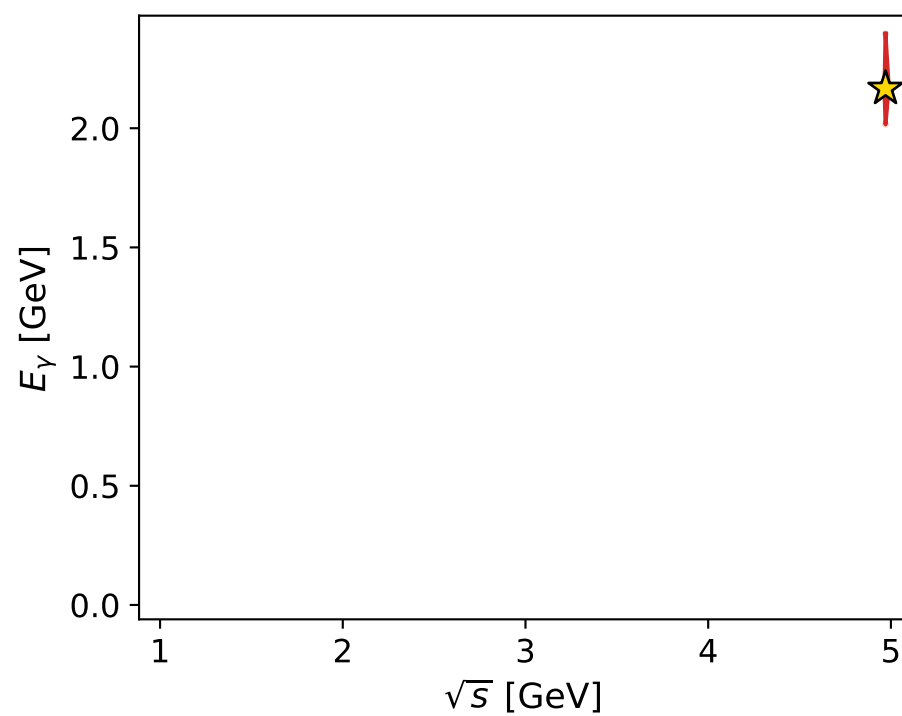
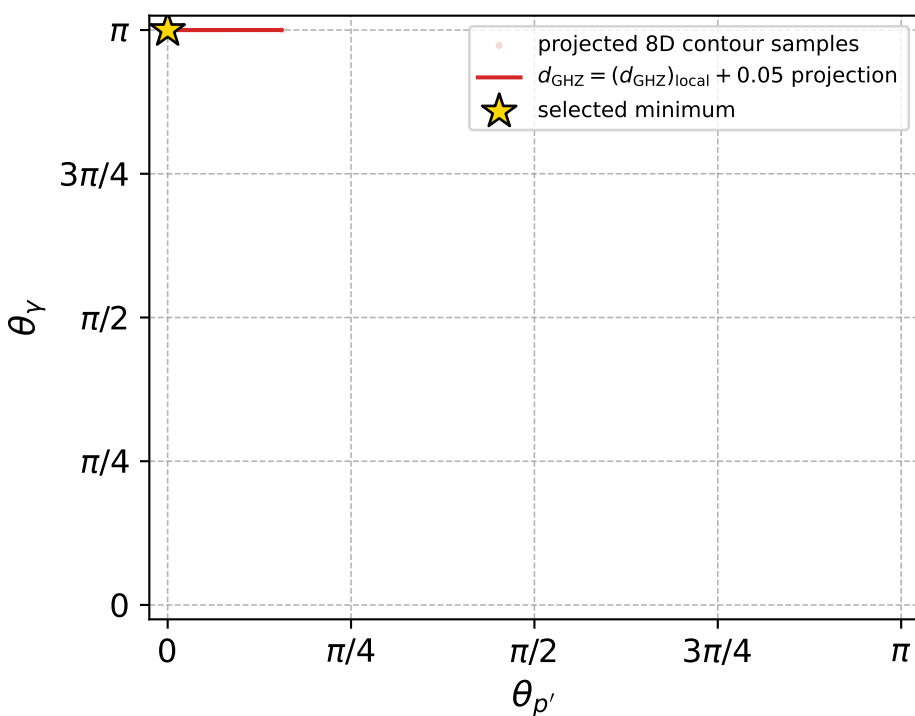
x--y plane



Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 0; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.2283462 \times 10^{-8}$
 $d_{\text{GHZ}} \text{ contour} = 0.050000022$
 above global minimum = 0
 8D contour samples = 415
 configured 8D directions = 1536

selected phase-space point:

$\text{sqrt_s} = 4.970528$
 $\text{theta_p_out} = 1.301108 \times 10^{-7}$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 2.166672$
 $\text{phi_p_out} = 1.742183$
 $\text{phi_gamma_out} = 3.74627$
 $\text{alpha_e} = 0.7579928$
 $\text{alpha_p} = 2.383596$